Climate Shocks, Health Finance, and Economic Outcomes

Main Results Recap + April 2026 Committee Feedback Response

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Roadmap

Part 1: where we were before April. Part 2: what changed in response to committee feedback.

Part 1 — Main Results Recap

- Project setup and empirical specification
- State-level FE: cold and heat drive opposite financial channels
- County-level FE: cold raises debt + premiums; heat depresses income
- Weather-swing (delta) analysis: AQI ratchet, drought-PCPI loss
- Cross-level synthesis

Part 2 — Committee Feedback Response

- Random-effects robustness (Hausman)
- Post-exit local-projection dynamics
- DiD with never-exposed counties (2012 drought; Callaway-Sant'Anna)
- Propagation pathways: literature + descriptives
- Humidity (PRISM): parked status
- What changed substantively; open caveats

Phases 0–6 documented; Phase 4 humidity parked.

What We Learned

Headline findings with strength-of-evidence

Finding	What it means	How we know	Strength
Cold raises household financial burden	Premiums, medical debt, hospital bad debt all rise with cold exposure	State FE, County FE, Δ-swing — three levels agree	Strong
Drought hits households through income	Income falls 1–2 years out $\rightarrow$ medical debt follows	State FE, County FE, 2012 natural-experiment DiD all converge	**Strongest**
Cold scarring shows up only across designs	Within-county recovery on exit (LP); ever-exposed populations end up worse off (DiD)	Post-Exit LP + Callaway-Sant'Anna with never-treated controls	Strong
Heat depresses household income	Med HH income falls with heat exposure (county-only)	County FE, Δ-swing	Moderate
Weather *volatility* carries costs beyond levels	AQI swings produce a debt ratchet level specs missed	Δ-swing LP — volatility identifies what levels can't	Strong
FE is the right specification	Random effects rejected for every estimable outcome	Hausman test (all $p < 10^{-9}$)	Conclusive

Part 1 — Main Results Recap

Project setup, state, county, weather-swing

Project Background

Climate, health finance, and economic outcomes in the U.S., 2011–2023

Research Question

- Do climate shocks (cold, heat, drought, air quality) affect the **cost and accessibility** of health care?
- Do they propagate into **household economic outcomes** — income, debt, employment?

Motivation

- Climate adaptation policy needs **health-finance channel** evidence
- Literature centres on mortality; **cost & debt channels understudied**
- U.S. two-way FE enables within-unit identification at state and county

Data Coverage

Domain	Source
Climate (Temp, HDD, CDD)	NOAA/NCEI
Drought index (PDSI)	NOAA/NCEI
Air quality (AQI)	EPA
HIX premiums	HIX Compare
Medical debt	Urban Institute
Hospital bad debt	NASHP HCT
Employer insurance	AHRQ MEPS-IC
Public payers	CMS NHE
HH income / employment	ACS / BEA

2011–2023; ∼3,155 counties; 51 states × 13 years.

Empirical Specification

Two-way fixed effects with distributed lags; complementary LP framework for dynamics; dollar outcomes inflation-adjusted to $2023

Dynamic Panel / Distributed Lag (DP/DL)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{h=0}^{2} \beta_h \, \mathrm{Shock}_{i,t-h} + \mathbf{X}_{it}'\delta + \varepsilon_{it}$$

- Main FE estimator for contemporaneous and lagged effects
- α_i unit FE (State or fips); γ_t year FE
- SE clustered at **state level**; RA cluster for premiums

Local Projections (LP, Jordà 2005)

$$Y_{i,t+h} = \alpha_i^h + \gamma_t^h + \beta_h \, \mathrm{Shock}_{i,t} + \mathbf{X}_{it}'\delta_h + \varepsilon_{i,t+h}$$

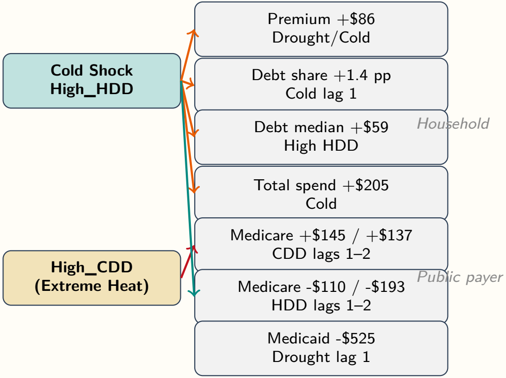
- Separate regression per horizon $h \in \{-2, \ldots, +3\}$
- Negative horizons $=$ placebo pre-trend checks
- More flexible if lag shapes are misspecified

Shocks: High_CDD, High_HDD, Z_Temp, Z_Precip, PDSI, High_AQI_Max ($>$100). **Outcomes:** employer premium contribution (MEPS-IC), benchmark silver premium (HIX), medical debt share/median (Urban), hospital bad debt per capita (NASHP), total/Medicaid/Medicare per-enrollee spending (CMS), HH income, employment, PCPI (ACS/BEA).

State-Level Findings

Cold raises premiums and debt; drought hits Medicaid and PCPI; heat and cold move public payers in opposite directions

State-Level: Cold and Heat Drive Opposite Financial Channels



Cold channel:

- Premiums, debt, and total spending rise
- Medicare falls with lagged HDD exposure

Heat channel (CDD):

- Medicare rises 1–2 years later

Drought:

- Medicaid falls and premiums rise

Heat and cold affect public payers in opposite directions — novel finding.

State: Headline Coefficients

Selected significant lags; $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$

Household outcomes

Coefficient (state-clustered)	Estimate	
Employer Prem.: Severe Drought ($h=0$)	+\$86.4	***
Employer Prem.: Severe Drought (lag 1)	+\$65.1	**
Employer Prem.: Pct $PM_{2.5}$ days	+\$20.8	***
Debt Share: Cold Shock (lag 1)	+0.0135	**
Debt Median: High HDD ($h=0$)	+\$58.9	**
Debt Median: Heat Shock (lag 2)	+\$47.7	**
Total PC Health Exp: Cold Shock ($h=0$)	+\$205.3	**

Public payer outcomes

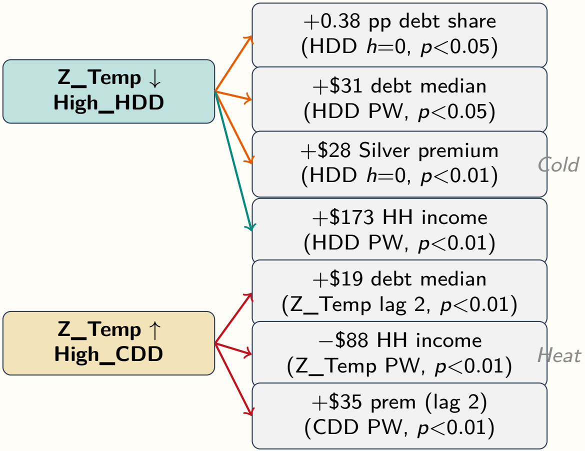
Coefficient (state-clustered)	Estimate	
Medicaid PE: Extreme Drought ($h=0$)	−\$566.2	**
Medicaid PE: Severe Drought (lag 1)	−\$524.5	***
Medicare PE: Severe Drought (lag 1)	−\$144.3	***
Medicare PE: High CDD (lag 1)	+\$144.7	**
Medicare PE: High CDD (lag 2)	+\$137.2	***
Medicare PE: High HDD (lag 1)	−\$110.1	*
Medicare PE: High HDD (lag 2)	−\$192.7	***

Headline: Extreme Drought lag-2 and Cold Shock lag-1 raise medical debt & premiums; drought depresses Medicaid; heat and cold move Medicare in opposite directions.

County-Level Findings

Cold raises debt and premiums; heat depresses income; hospital bad debt rises with cold and lagged drought

County-Level: Channels Mirror the State Findings



Cold channel:

- Raises debt prevalence & severity
- Raises premiums on impact
- Raises HH income (energy mix)

Heat channel:

- Raises debt median (lagged)
- **Reduces** HH income contemporaneously
- Raises premiums with 1–2 yr lag

Cross-level alignment:

- Cold → debt: both levels confirm
- Heat → income: county only
- Premium channels align in direction

County: Headline Coefficients and Hospital Channel

Two-way FE (fips + Year); state-clustered (RA-cluster for premiums); **$p<0.05$, ***$p<0.01$

Premiums, debt, income (PW)

Coefficient	Estimate	
Benchmark Silver: High HDD (h=0)	+$21.9	***
Benchmark Silver: High CDD (lag 2)	+$35.3	***
Debt Median (PW): High HDD (h=0)	+$31.2	**
Med HH Income: Z_Temp (h=0)	−$88	***
Med HH Income: High HDD (h=0)	+$173	***
PCPI: PDSI lag 2	−$113	**
PCPI: RA-spec PDSI (h=0)	−$206	***

Hospital bad debt per capita

Coefficient	Estimate	
High HDD (h=0)	+$5.33	***
Z_Temp lag 2 (UW)	−$2.41	***
Z_Temp lag 2 (PW)	−$2.08	***
PDSI lag 2 (PW)	+$0.50	*
Uninsured Rate (control)	+$151	***

Cold raises hospital bad debt on impact; heat lag *reduces* it; uninsured rate dominates.

Weather-Swing (Delta) Analysis

Year-over-year climate volatility as a new source of variation;
new findings the level specs missed

Weather Swings: AQI Ratchet, Drought-PCPI, Temp-Debt

ΔFE primary spec; LP horizons $h = 0 \ldots 3$; $^{*}p{<}0.10$, $^{**}p{<}0.05$, $^{***}p{<}0.01$

Selected swing coefficients

Coefficient	Estimate	
ΔHDD → Silver Prem (h=0)	+0.033	***
HDD Onset → Silver Prem	+$49.5	**
ΔZ_Temp → Debt Median (h=3)	+$29.7	***
ΔMax_AQI → Hosp Bad Debt (h=0)	+0.005	**
ΔMax_AQI → Hosp Bad Debt (h=3)	+0.016	***
ΔZ_Precip → PCPI (h=1)	−$274	***
ΔPDSI → PCPI (h=1)	−$226	***

What the delta analysis adds

- **AQI ratchet** (h=3 effect 3× h=0) — level specs were null; volatility identifies cost
- **Drought worsening** drives PCPI loss; recovery asymmetrically weaker
- **Temp swing** accumulates into medical debt over 3 years
- **HDD onset pricing**: first-year cold raises Silver Premium ∼$49

Key insight: weather *volatility* carries independent health-finance costs beyond level effects.

Cross-Level Synthesis

State, county, and weather-swing analyses tell a consistent story

Finding	State	County	Swing
Cold / HDD $\rightarrow$ medical debt $\uparrow$	✓	✓	✓
Cold / HDD $\rightarrow$ premiums $\uparrow$	✓	✓	✓
Cold / HDD $\rightarrow$ hospital bad debt $\uparrow$	—	✓	—
Drought $\rightarrow$ Medicaid $\downarrow$ / PCPI $\downarrow$	✓	✓	✓
Heat / CDD $\rightarrow$ HH income $\downarrow$ (lag)	—	✓	(asym.)
AQI $\rightarrow$ hospital bad debt $\uparrow$	(weak)	(weak)	✓ (new)
Heat vs. cold on *Medicare* (opposite signs)	✓	(partial)	—

Headline: cold shocks raise household financial burden; drought hits public payers and PCPI; weather *volatility* adds an independent channel (AQI ratchet; temp-debt accumulation).

Part 2 — Committee Feedback Response

Random effects, post-exit dynamics, never-exposed DiD, propagation evidence, humidity (April 2026 review)

What the Committee Asked

Five concerns across three domains, all addressed in this update except humidity

Econometric

- **Check random effects** as an alternative to FE
- **Lag / exit tests**: what happens when a county leaves a shock state?
- Find **natural experiments** for DiD — are there never-exposed counties?

General

- **Propagation pathways** must be backed by evidence (heat $\rightarrow$ delayed care; cold $\rightarrow$ utilization shifts)

Environmental

- Add **humidity** (PRISM `tdmean`) at state level

Response status

Concern	Status
Random effects	Done
Post-exit dynamics	Done
Never-exposed DiD	Done
Propagation pathways	Done
Humidity (PRISM)	**Parked**

Random Effects Is Rejected; Headlines Survive

Hausman tests on primary FE specs; matched plm two-way effects

Design

- Two-way FE vs. RE GLS on the same RHS; Hausman test per outcome
- 11 outcomes estimable (4 state, 7 county)

Verdict

- **RE rejected for every estimable outcome at $p < 10^{-9}$**
- Headlines (Drought lag-2, Cold lag-1) have the **same sign** in RE
- Only Medicaid Per Enrollee shows material FE/RE divergence ($\sim 2\times$)

Selected Hausman results

Outcome	χ^2	p-value
State		
Medical Debt Share	233.6	4.5e-40
Total PC Health Exp	247.9	5.7e-43
Medicaid PE Health Exp	81.3	2.3e-10
Medicare PE Health Exp	834.4	2.1e-166
County		
Medical Debt Share	23,905	<1e-300
Benchmark Silver Real	23,545	<1e-300
PCPI Real	5,809	<1e-300
Civilian Employed	220.2	4.6e-41

Post-Exit Dynamics — What Happens When a County Leaves Shock State?

Local projections of $*$_Exit at $h = 0, 1, 2, 3$; answers the committee's exit-test question directly

Drought_Exit

- **PCPI partial scarring**: $+\$1{,}044$ p.c. at $h = 2$ ($p = 0.0002$)
- Recovery rebounds but does not fully restore pre-shock income
- Confirmed by Block B interaction ($+\$964$ at $h = 2$, $p = 0.010$)

HDD_Exit

- **Immediate hospital-cost relief**: -3.13 p.c. at $h = 0$ ($p = 0.014$)
- Sustained at $h = 2$: -2.88 ($p = 0.034$)
- Cold-shock cost is in the **exposure**, not lasting damage

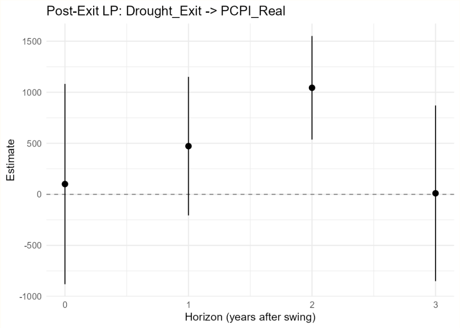
CDD_Exit

- Persistent employment benefits after exit
- Likely part secular regional trend (caveated)
- Phase 2 LP rejects within-county heat scarring

Post-Exit LP: Shock-Specific Recovery Patterns

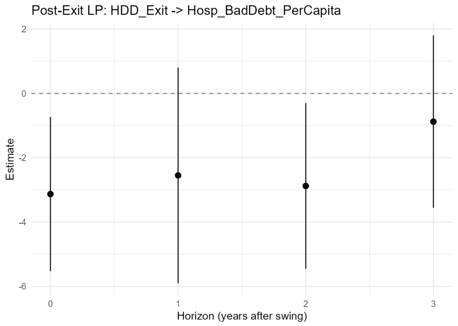
Drought partially scars income; cold-shock exit brings immediate hospital-cost relief

Drought_Exit → PCPI Real



Peak at $h = 2$: $+\$1{,}044$ ($p = 0.0002$). Partial recovery.

HDD_Exit → Hosp. Bad Debt / Capita



Relief at $h = 0$ (-3.13, $p = 0.014$); sustained at $h = 2$. No scarring.

Phase 0: Are There Never-Exposed Counties?

Inventory of zero-event counties over 2011–2023; feasibility memo selected primary cohorts

Never-exposed pool by shock

Shock	Total	Never-exposed	Share
Extreme Drought (PDSI ≤ -4)	3,225	**2,534**	78.6%
High HDD (top quintile)	3,225	**2,303**	71.4%
High CDD (top quintile)	3,225	**2,121**	65.8%
High AQI Max ($>$100)	3,225	100	**3.1%**

- Drought, HDD, CDD all have large never-exposed pools
- **AQI dropped from DiD**: 3.1% control pool too thin

Candidate sharp events

Event	Onsets	Onset rate
2012 Midwest Drought	139	4.4%
2022 Drought	110	3.5%
2013 HDD (cold)	407	13.0%
2018 CDD (heat)	250	8.0%
2023 AQI (wildfire smoke)	336	34.1%

2012 Midwest Drought: The Cleanest Natural Experiment

139 first-event counties (treatment) vs. 2,534 never-exposed (control), 2011–2023

Design

$$Y_{it} = \alpha_i + \gamma_t + \tau\,(\text{Treated}_i \times \text{Post}_t) + \varepsilon_{it}$$

- Treated: first-ever Extreme Drought event = 2012
- Control: never experienced Extreme Drought in panel window
- Cluster: state; SE: `feols(..., cluster="State")`

Why this is the cleanest piece

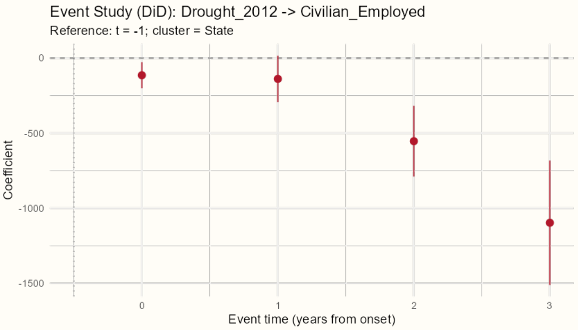
- Within-county LP cannot rule out recurring-shock confounding
- 2x2 against never-exposed **closes that loophole**
- 2012 drought is a documented event with defined geography

2x2 ATT results

Outcome	ATT	p
Medical Debt Share	−0.0062	0.49
Hosp Bad Debt PC	+6.26	0.29
PCPI Real	−\$1,311	**0.027**
Med HH Income Real	−\$992	0.24
Civilian Employed	−2,053	**0.0001**
Benchmark Silver Real	—	ACA collinearity

2012 Drought DiD: Employment Loss Accumulates Over the Event Window

Event-study with leads/lags; reference $t = -1$; cluster = State; treated vs. never-exposed only



Reading the figure

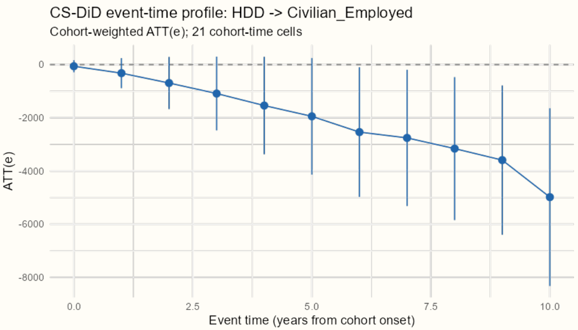
- $t = 0$: year of first drought event (2012)
- Visible **near-zero pre-period** at $t = 0$ (only 1 pre-year given panel start in 2011)
- Post-event employment loss **accumulates**: $\sim$-100 at $t = 1$; $\sim$-550 at $t = 2$; $\sim$-1,100 at $t = 3$
- Consistent with the income-channel pathway feeding into employment

Same identification, different outcome: PCPI Real follows the same downward profile (peak ATT $-\$1{,}311$).

Callaway-Sant'Anna: Long-Run Cold-State Scarring

Cohort-weighted ATT(e) on HDD with never-treated controls; long-run damage compounds over a decade

HDD $\rightarrow$ Civilian_Employed (CS-DiD)



Cumulative loss grows from $\sim$0 at $e = 0$ to $-4{,}982$ **at** $e = 10$ ($p = 0.003$). Medical Debt Share +4.9pp at $e = 10$.

Drought (4 cohorts, 268 treated)

- PCPI at $e = 0$: $-\$1{,}050$ ($p = 0.002$)
- Civilian Employed at $e = 11$: $-3{,}915$ ($p = 0.0001$)

HDD (2 cohorts, 230 treated)

- Long-run scarring in employment and debt-share
- Cumulative through $e = 6$–10

CDD: noisy, regional confounds

- Treatment concentrated in TX/GA/MS/FL; controls in north
- Signs reverse across horizons — **de-emphasize**

LP and DiD Are Complementary, Not Contradictory

The cold-scarring puzzle resolves once we see both designs measure different things

Local Projections / Exit-LP

- Estimand: within-county response when shock state changes year-over-year
- HDD_Exit $\rightarrow$ hospital-cost relief at $h = 0$
- **Read:** no within-county cold scarring after recovery

CS-DiD with never-exposed

- Estimand: persistent gap between ever-treated and never-treated populations
- HDD long-run: $-4{,}982$ jobs and $+4.9$pp debt share at $e = 10$
- **Read:** treated counties end up worse off long-run

Reconciliation: cold-shock counties *recover* when shocks end (LP), but they *spend more years in shock state* than never-exposed counties do, so the **population gap accumulates** (CS-DiD). Both are correct; they measure different things.

Propagation Pathways — Now Evidence-Backed

Each headline channel anchored to 2–4 peer-reviewed references with direction and time-scale

Pathway	Time-scale claim	Where identified	Strength
Heat $\rightarrow$ delayed care	Within-year deferral; lag-1/2 utilization	County Spec 2, CS-DiD CDD	Moderate
Cold $\rightarrow$ utilization shift	Contemporaneous + lag-1; long-run scarring at $e \geq 8$	State cold-shock lag-1, CS-DiD HDD	Strong
Drought $\rightarrow$ income $\rightarrow$ debt	Lag-1, lag-2 income; debt follows	State drought lag-2, 2012 DiD, Exit-LP recovery	**Strongest**
AQI $\rightarrow$ respiratory / cardiac	Within-year, episodic spikes	County continuous AQI, event-study, swing analysis	Limited

Refs **Drought:** Hornbeck (2012); Burke-Hsiang-Miguel (2015); Carleton et al. (2022). **Cold:** Deschênes-Moretti (2009); Gasparrini et al. (2015); Andrews et al. (2017). **Heat:** Sun et al. (2021); White (2017); Karlsson-Ziebarth (2018). **AQI:** Deryugina et al. (2019); Schlenker-Walker (2016); Currie-Walker (2011).

Humidity Integration — Status and Resumption Plan

Phase 4 deferred; the highest-priority residual gap from committee feedback

What we know

- PRISM gridded endpoint returns monthly BIL rasters
- No state-aggregate API; must aggregate locally
- ∼408 monthly rasters, ∼1.5 GB on disk

Blocker

- Raster → state polygon aggregation needs three R spatial packages
- Not installed in local R 4.2.2; Windows install pulls heavy system deps (GDAL/GEOS/PROJ)

Resumption plan

- Install spatial stack; download monthly tdmean
- Aggregate to state-year; join into master
- Re-run; test **drought lag-2** and **cold lag-1** survival

Why this matters

- Single residual gap from committee feedback
- Low-humidity confounding remains open

The Analysis Now Has Two Pieces of Stronger Identification

And one genuine puzzle resolved by treating LP and DiD as complementary

New evidence

- **2012 Drought 2x2** is now the cleanest single identification piece (PCPI −\$1,311; Employed −2,053)
- **HDD long-run scarring** (−4,982 jobs at $e = 10$; +4.9pp debt share at $e = 10$) was not visible in the within-county LP framework
- Drought_Exit shows **partial income scarring** — recovery rebounds but does not fully restore

Resolved puzzle

- Cold counties **recover on exit** (no within-county scarring)
- But the ever-treated population **ends up worse off** long-run (DiD)
- Both findings are correct; they measure different things

Methodological closure

- Random effects rejected; FE is the right spec
- AQI exclusion from DiD now defensible (3.1% never-exposed)

Open Caveats and Residual Risks

What the committee should know is still not addressed

Humidity confounding

- PRISM tdmean not yet a control
- Headline drought lag-2 and cold lag-1 **not stress-tested** against humidity
- Mitigation: resumption plan in place

CDD CS-DiD region confound

- Treatment concentrated south; controls north
- Two-way FE leaves regional trends
- Reported as **suggestive only**

Multiple testing

- 427 ATT(g,t) cells $\times$ 7 outcomes $\times$ up to 11 event-times
- Headline cells survive Bonferroni within-shock
- Smaller-cohort cells do not

County-master deduping

- Master not strict one-row-per-(fips, year)
- Deduped in every analysis script
- Underlying fix deferred

Thank you.

Discussion and questions